

Project Design Phase Proposed Solution

Date	05 July 2026
Team ID	
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	2 Marks

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Grid planners, policy analysts, and industrial consumers rely on scattered, unaggregated state-wise daily consumption spreadsheets with no forecasting or regional comparison tools — making them reactive to demand spikes rather than proactive, and slowing down renewable-investment decisions.
2.	Idea / Solution description	A web-based Electricity Consumption Analytics Platform that automatically ingests and cleans state-wise daily usage data, visualizes it through real-time regional dashboards and geographic heatmaps, generates short-term demand forecasts, and sends threshold-based peak-load alerts — all from a single, trusted interface.
3.	Novelty / Uniqueness	Unlike generic BI tools (Power BI/Tableau) that require manual setup for every report, this platform is purpose-built for the power sector: it combines automated ingestion, region-level (NR/WR/SR/ER/NER) comparison, and demand forecasting with proactive alerting in one workflow, instead of three separate manual steps.
4.	Social Impact / Customer Satisfaction	Better demand forecasting reduces the risk of blackouts and load-shedding for the public; faster, clearer insight helps planners direct renewable infrastructure to the states that need it most; and grid operators spend less time on manual spreadsheet work and more time on proactive decision-making.
5.	Business Model (Revenue Model)	SaaS subscription tiered by number of states/regions monitored, sold to state electricity boards and central energy ministries; an additional premium tier offers advanced forecasting models and API access for integration with third-party grid management systems.
6.	Scalability of the Solution	The microservices architecture (ingestion, forecasting, alerting) allows new states, regions, or data sources to be added without redesigning the platform, and containerized deployment (Docker/Kubernetes) lets compute scale independently as more states or higher-frequency data are onboarded.